

E - Geometric Progression

Editorial (/contests/abc293/tasks/abc293_e/editorial)

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 500 points

Problem Statement

Given integers A , X , and M , find $\sum_{i=0}^{X-1} A^i$, modulo M .

Constraints

- $1 \leq A, M \leq 10^9$
- $1 \leq X \leq 10^{12}$
- All values in the input are integers.

Input

The input is given from Standard Input in the following format:

A X M

Output

Print the answer.

Sample Input 1

Copy

3 4 7

Copy

Sample Output 1

Copy

5

Copy

$3^0 + 3^1 + 3^2 + 3^3 = 40$, which equals 5 modulo 7, so 5 should be printed.

Sample Input 2

Copy

8 10 9

Copy

Sample Output 2

Copy

0

Copy

Sample Input 3

Copy

1000000000 1000000000000 998244353

Copy

Sample Output 3

Copy

919667211

Copy